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Nature: short note — 一条初等引理 + 对已发表定理的常数改进（不含新猜想的证明）

Source theorem: Palojärvi, arXiv:1807.01506v3, Theorem 4.1

Archive basis: docs/E4-ENGINE-2-elementary-covering-lemma-selfcontained-m1-and-better-constant.md

§0 Abstract

We record an elementary covering lemma: for every $z\in\mathbb{C}$ with $|z|=1$,

$$\max_1\leq k\leq s\operatorname{Re},zk\geq (1)/(2) .$$

We use it to give a self-contained proof of the m=1 case of Palojärvi's Theorem 4.1, replacing the invoked large-sieve-type bound (constant 1/20, Montgomery, *Ten Lectures*, Ch. 5 Thm 11 = Palojärvi Lemma 2.2) by an elementary constant 1/2. The substitution improves the detection threshold: the coefficient of $(\operatorname{KF}_{,1}(\tau)+\operatorname{KF}_{,4}(\tau))$ becomes exactly one tenth of the source's, and the source's $20,n\log n$ main term disappears.

§1 The lemma

Lemma C. Let $z\in\mathbb{C}$, $|z|=1$. Then

$$\max_1\leq k\leq s\operatorname{Re},zk\geq (1)/(2) .$$

Proof. Write $z=ei\theta$, so $\operatorname{Re},zk=\cos k\theta$. It suffices to show that

$$\bigcup_{k=1}^s\bigcup_{j\in\mathbb{Z}}\frac{1}{k}(360^\circ+j+[-60^\circ,60^\circ])=\mathbb{R}\bmod{360^\circ},$$

because θ lying in the set for some k means $|k\theta|\leq 60^\circ\pm\text{od}360^\circ$, hence $\cos k\theta\geq 1/2$. The intervals (in degrees) are

$$k=1:[0,60]\cup[300,360];\;k=2:[150,210];\;k=3:[100,140]\cup[220,260];$$

$$k=4:[75,105]\cup[165,195]\cup[255,285];\;k=5:[60,84]\cup[132,156]\cup[204,228]\cup[276,300].$$

They cover the circle consecutively:

$$\begin{aligned} &[0,60]_{(1)}\rightarrow[60,84]_{(5)}\rightarrow[75,105]_{(4)}\rightarrow[100,140]_{(3)}\rightarrow[132,156]_{(5)}\rightarrow[150,210]_{(2)}\\ &\rightarrow[165,195]_{(4)}\rightarrow[204,228]_{(5)}\rightarrow[220,260]_{(3)}\rightarrow[255,285]_{(4)}\rightarrow[276,300]_{(5)}\rightarrow[300,360]_{(1)} . \end{aligned}$$

square

Numerical check (复核): minimizing numerically over a fine grid gives $\min\theta\max_1\leq k\leq s\cos k\theta=0.5$, attained at $\theta=60^\circ$ and 300° (finest-grid value 0.5000014).

Optimality. The constant 1/2 is best possible: at $\theta=60^\circ$, $(\cos k\theta)_{k=1}^5=(1/2,-1/2,-1,-1/2,1/2)$, so the maximum equals exactly 1/2. Hence no larger constant can hold uniformly.

Comparison with the source's engine [出处]: the source invokes $\max_1\leq n\leq sM\operatorname{Re}\overline{\sum}\leq Mzj^n\geq (1)/(20)$; at M=1 Lemma C gives $(1)/(2)>(1)/(20)$, i.e. ten times stronger *and* elementary (no large-sieve input).

§2 Application: Palojärvi Theorem 4.1 at m=1

Source statement [出处] (Palojärvi, Thm 4.1): let F satisfy (a)–(d), $\tau>1/e$, and suppose F has at most one zero ρ_1 with $|\rho_1/(\rho_1-\tau)|>1$. Then

$$\rho_1\text{ exists}\iff |\operatorname{Re},\lambda F(n,\tau)|\geq (\operatorname{KF}_{,1}(\tau)+\operatorname{KF}_{,4}(\tau)),n\log n\text{ for some }n\in[N,5N],\;N\mid n .$$

Decomposition (source (37), verbatim [出处]): with $w\rho=\rho/(\rho-\tau)$,

$$\operatorname{Re},\lambda F(n,\tau)=\underbrace{\lim_{t\rightarrow\infty}\sum_{T(n)<|\operatorname{Im}p|\leq t,\;0\leq\operatorname{Re}p\leq\tau/2}\operatorname{Re}(1-w\rho^n)(i)}_{\text{large height}}+\sum\cdots(ii)+\sum\cdots(iii),$$

where $T(n)=n\tau$ and $|w\rho|\leq 1\iff \operatorname{Re}p\leq\tau/2$.

- (Leftarrow) no off-axis zero. Then term (iii) is absent and every zero in (i)(ii) satisfies $|w\rho|\leq 1$, so $|\operatorname{Re}(1-w^n)|\leq|1-w^n|\leq 1+|w|^n\leq 2$. By the source's Theorem 2.1,

$$|\operatorname{Re},\lambda F(n,\tau)|\leq (\operatorname{KF}_{,1}+\operatorname{KF}_{,4})n\log n\;(\forall n),$$

with the same constants as the source (using $|1-w^n|$ loses no constant).

- (Rightarrow) an off-axis zero ρ_1 with $R'\mathrel{:=}|w_1|\geq R>1$. Put $z\mathrel{:=}w_1N/R'$, so $|z|=1$. By Lemma C choose $k\leq 5$ with $\operatorname{Re},zk\geq(1)/(2)$; then

$$\operatorname{Re}(1-w_1^n)=1-R'^n,\operatorname{Re},zk\leq 1-(1)/(2)\;R'^n\;(R'^n\geq R^n),$$

hence

$$|\operatorname{Re},\lambda F(n,\tau)|\geq (1)/(2)\;R'^n1-(\operatorname{KF}_{,1}+\operatorname{KF}_{,4})n\log n\geq (\operatorname{KF}_{,1}+\operatorname{KF}_{,4})n\log n$$

as soon as

$$\boxed{R'^n\geq 4(\operatorname{KF}_{,1}(\tau)+\operatorname{KF}_{,4}(\tau)),n\log n+2},$$

Comparison with the source [出处·已逐字核](2026-09-18): the source's threshold is

Verbatim (source, p.20): "It is sufficient to show $R^n\geq 40n\log n((1)/(2)+\operatorname{KF}_{,1}(\tau)+\operatorname{KF}_{,4}(\tau))$."

$$R'^n\geq 40,n\log n((1)/(2)+\operatorname{KF}_{,1}+\operatorname{KF}_{,4})=20,n\log n+40(\operatorname{KF}_{,1}+\operatorname{KF}_{,4})n\log n,$$

so

$$[20n\log n+40(\operatorname{KF}_{,1}+\operatorname{KF}_{,4})n\log n]-[4(\operatorname{KF}_{,1}+\operatorname{KF}_{,4})n\log n+2]=20n\log n+36(\operatorname{KF}_{,1}+\operatorname{KF}_{,4})n\log n-2>0$$

(for $n \geq e$). Thus in the same window and with the same conclusion the threshold is strictly smaller: the $(K_{F,1}+K_{F,4})$ coefficient drops to $1/10$ of the source's, and the $20n \log n$ main term is additionally saved.

Explicit N. Replacing (1)/(20) by (1)/(2) in the three-scale split of the source's §5b gives

$$N_{\text{new}} = \lceil \max(T_0) / (e\tau), \exp(-W_{-1}(-(2)/(3)\log R)), (12\log(4(K_{F,1}+K_{F,4})+2)) / (\log R) \rceil,$$

i.e. the $(12)/(\log R)$ slot is unchanged and only the constant $40(K_{F,1}+K_{F,4})+20$ is replaced by $4(K_{F,1}+K_{F,4})+2$; the m-cost in N decreases accordingly.

§3 The case $m \geq 2$: still one external dependence

- (a) Fejér-weight route 【严格】 (self-contained, but needs a **comparability** assumption): with Fejér weights $w_k = 1 - k/(5M)$,

$$\sum_{k \leq s} M w_k |\sum_j z_j k|^2 \geq \sum_j D_j \text{-frac}{M(M-1)} 2, \quad D_j := \sum_{k \leq s} M w_k |z_j|^2 k,$$

(diagonal ≥ 0 ; each off-diagonal pair $\geq -(1)/(2)$ by non-negativity of the Fejér kernel), hence

$$\max_{k \leq s} M |\sum_j z_j k|^2 \geq (\sum_j D_j - M(M-1)/2) / (5M/2).$$

Numerically 【复核】: if all $|z_j|=1$, $\sum_j D_j \approx (5M/2)M$ gives the bound $\geq (4)/(5) M$, i.e. $\geq \sqrt{0.8}M$. Limitation: it needs $\sum_j D_j > M(M-1)/2$, i.e. most exceptional moduli close to the maximum (comparability); for widely spread moduli the route degenerates.

- (b) Still Montgomery: the **real-part** detection at $m \geq 2$ with spread moduli still needs Lemma 2.2; we did not make it self-contained here.

§4 Scope and honesty

- Claimed: Lemma C (elementary, complete proof, constant optimal) and, via it, a self-contained proof of the $m=1$ case of Pálójärvi's Theorem 4.1 with a strictly better threshold $((K_{F,1}+K_{F,4})$ -coefficient reduced by a factor 10; the $20n \log n$ term removed).

- Not claimed: any progress on the $m \geq 2$ case; any new equivalence criterion; any RH-related statement. The source theorem's **structure** is used as-is (【出处】), including its decomposition (37) and Theorem 2.1.

- 【✓】 【出处·已逐字核】 (2026-09-18): the source's threshold formula, including its $20n \log n$ term, is verbatim confirmed against Pálójärvi arXiv:1807.01506v3 (p.20). The core comparison of this note therefore rests on verified source text, not on an archive transcription.

- No RH is used anywhere in this note; the input is the source theorem plus elementary covering.

§5 Positioning and literature check (2026-09-18)

Searched (web, 2026-09-18): single-unit-circle power sums, Turán power-sum lemma, covering of the circle by arcs, optimal constants.

Found — Lemma C sits inside the classical power-sum (Turán) family:

- Turán's method gives lower bounds for exponential/complex power sums; it was introduced by Turán (1947) in connection with the zeros of the Riemann zeta function 【外搜】.

- Known **multi-point** statements: for $z_1, \dots, z_n$ on the unit circle, $\max_1 \leq \text{ell} \leq n |\sum_{j=1}^n z_j^{\text{ell}}| \geq 1$ (equality iff the z_j are the vertices of a regular $(n+1)$ -gon) 【外搜】; Cassels' lemma ($b_j > 0$, $|z_j|=1$, $K > N$): $\max_k \leq K |\sum| \leq N b_j |z_j| \geq \sqrt{(K-N)/K, (\sum b_j^2)^{1/2}}$ 【外搜】.

- Turán's second theorem: for $|z_n| \geq 1$, any v -range of length N contains some v with $|sv| \geq c(M,N)|s_0|$ 【外搜】.

- The source's own engine is the M -point quantitative form (Montgomery, **Ten Lectures**, Ch. 5 Thm 11 = Pálójärvi Lemma 2.2), with constant $1/20$ over $1 \leq n \leq 5M$.

Not found: the single-term case ($M=1$) in **optimal** form — i.e. $\max_1 \leq k \leq s \text{Re}, z_k \geq (1)/(2)$ with an elementary covering proof — stated anywhere. $\triangle$ This is a search-level statement, not a claim that it is absent from the literature; the lemma is elementary and is a specialization of a classical family, so it may well be folklore.

Therefore this note's claimable content is:

- the elementary optimal-constant lemma, recorded with a short self-contained proof (possibly folklore; presented as a tool, not as a novelty claim), and
- ★ the application: in the $m=1$ case of a **published** theorem, the $(K_{F,1}+K_{F,4})$ -coefficient of the detection threshold is reduced by a factor 10 and the source's $20n \log n$ main term is removed — checked verbatim against the source (p. 20).

Recommendation: state Lemma C with a provenance note ("elementary; to our knowledge not recorded in optimal form"), and rest the note's claim on item 2.

References

- M. Pálójärvi, **On the Li criterion and the explicit zero-free region**, arXiv:1807.01506v3, Theorem 4.1 (and Lemma 2.2, Theorem 2.1).

- H. L. Montgomery, **Ten Lectures on the Interface between Analytic Number Theory and Harmonic Analysis**, Ch. 5, Thm 11 (the replaced engine).